

# Matthew Huynh

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## Education

University of California, Merced | Merced, CA  
Bachelor of Science in Mechanical Engineering (ME)  
Minor in Materials Science & Engineering (MSE)

Expected Graduation: December 2026  
GPA: 3.45

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## Skills

Mechanical Design: SolidWorks, Inventor, Onshape  
Electronics & Fabrication: KiCad, Soldering,  
Breadboarding, Additive Manufacturing

Controls: MATLAB, Simulink  
Programming: Python

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## Experience

Mechatronics, Embedded Systems & Automation (MESA) Lab | Atwater, CA

June 2025 - Present

### *Undergraduate Research Assistant*

- Co-designed a multirotor UAV platform, performing thrust-to-weight, payload, power, and endurance calculations to guide motor, propeller, ESC, battery, and airframe selection.
- Supported integration of UAV and repurposed UGV platforms for autonomous gas-source localization, including early ROS-based coordination and preparation for physical testing.
- Fabricated custom motor mounts and supported mechanical/electrical integration as long-lead UAV components arrived.

Beutel Lab (Collaboration with MESA Lab) | Merced, CA

August 2025 - July 2026

### *Undergraduate Research Assistant*

- Reverse-engineered a Brooks Rand Model 3 mercury analyzer and contributed to subsystem architecture for a compact airborne sensing payload.
- Designed single- and multi-channel UV LED control electronics, evaluated PMT/socket interfaces, and validated control circuitry on a breadboard.
- Contributed to EMG-based signal fitting and preliminary mercury transport/detection modeling using experimental detector data.

Office of Research and Economic Development | Merced, CA

February 2023 - June 2025

### *Intellectual Property & Innovation Intern*

- Drafted patent applications with detailed CAD-based technical figures supporting engineering designs.
  - Coordinated technical documentation and data/material transfer agreements for research projects.
  - Collaborated with multidisciplinary teams to ensure technical accuracy and compliance.
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## Projects

### Flapping-Wing Ornithopter — In Progress

- Developing a lightweight flapping-wing mechanism in SolidWorks, integrating a geared drivetrain, crank-linkage system, and articulated wing structure.
- Developed a custom kinematic sizing tool for component sizing, interference, clearance, and tolerance analysis of laser-cut and 3D-printed parts.

### Self-Balancing Robot

- Designed and tuned a PID control system to stabilize a two-wheeled robot using IMU-based angle estimation.
- Implemented sensor fusion and iterative control tuning to improve system stability and dynamic response.

### CITRIS eVTOL Routing & Simulation System

- Led development of a graph-based eVTOL routing and operations simulator incorporating weather, airspace, population, hazard, and range constraints using Dijkstra-based path planning.
  - Built the FastAPI backend and React + Leaflet interface for fleet simulation and route visualization; designed operational state-machine logic for pre-flight, contingency, and post-flight handling.
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## Selected Publication

- Giri, S., Zhao, T., **Huynh, M.**, Chen, Y. "Information-Guided Safe Reinforcement Learning for Autonomous Gas Source Localization using sUAS." arXiv, 2026.

## Awards

- Best In Class - Excellent Team Project for Inclusive Innovation